

CS704 Project 2 (40%)

Overview

You have been employed by the AICorp to develop a prototype system to better monitor the status and safety of their employees within a factory. This will be accomplished through the use of an embedded device, with accelerometer and magnetometer, to track a user's step count and orientation. Following the AICorp standard procurement procedures, several designers (i.e. you) have been approached to undertake the project.

The AICorp has a wide variety of interests in personal and environmental monitoring. To streamline the development process, each contracted designer is being provided with a SensorTile development kit with a STLINK-V3 programmer. This kit is intended to be used as a mobile device, which is attached to a human user and track their movements in real-time. For prototyping purpose, the device can send data to a mobile app via Bluetooth to pass real-time location for visualisation.

Deliverables

1. Deliverable 1: Literature review report (5%), due 6th Oct 5pm (Canvas submission)

- **An individual literature review report** (up to 1 page excluding the references section, with the provided template) that contains a summary and comparison of different approaches of step and orientation detection using IMU-based sensors. In addition to the 2 references provided on Canvas, you **MUST** read at least another 4 papers and summarise/compare the approaches in your report.

2. Deliverable 2: Group interim lab demo (5%), due 6th Oct (Week 10 Lab Tuesday 2-4pm)

- You **MUST** demonstrate sending raw sensor (i.e. accelerometer and magnetometer) data to a mobile app via Bluetooth.
- You **MUST** demonstrate the ability to interpret sensor data value.

3. Deliverable 3: Group final demo and interview (15%), due 20th Oct (Week 12 Lab Tuesday 2-4pm)

- You **MUST** demonstrate tracking human motion (step count and orientation) on a given scenario, and send step count and orientation data to the provided mobile app via Bluetooth
- The tracking performance will be evaluated based on the step count and orientation accuracy during the different scenarios.
- The scenarios will be given at the interview.

4. Deliverable 4: Group code submission (3%), due 20th Oct 2pm (Canvas submission)

- The code should have a clear structure with well-commented implementation for gathering sensor data and compute real-time step count/orientation.
- The code should be zipped and named as **project2_groupXX.zip** (e.g. project2_group01.zip) for Canvas submission.
- Algorithms and software components **MUST** be modular and well documented
- Software components **MUST** be provided with compilation instructions in a top level README.md file
- Software components **MUST** be buildable in the STM32Cube IDE.

5. Deliverable 5: Group report submission (12%), due 19th Oct 5pm (Canvas submission)

- The report **MUST** be up to 4 pages with the provided template.
- The report **MUST** contain a description of the algorithms used.
- The report **MUST** contain performance analysis for the system.
- The report **MUST** contain discussion of the limitations of the system.
- The final report (as a PDF) should be named as **project2_groupXX.pdf** for Canvas submission.